



www.debmarineamibia.com



THE DEBMARINE NAMIBIA SAFETY MATURITY JOURNEY

4 August 2026

Chamber of Mines Safety Conference





6
Vessels

974
Employees

DEBMARINE NAMIBIA OPERATIONS: THE DIAMOND RECOVERY VESSELS

GEOPHYSICAL SURVEY



SAMPLING



SS Nujoma



SAFETY & SUSTAINABLE DEVELOPMENT



Debmar Pacific



!Gariep



Debmar Atlantic

Crawler vessels: ~65%



Mafuta



Benguela Gem



www.dedmaninamibia.com



A COMPLIANT CULTURE & REGULATIONS

A HIGHLY REGULATED AND COMPLIANT INDUSTRY

Namibia Law/regulation

Labour
Act, 11 of
2007

Merchant
Shipping
Act, 1951

Regulations relating
to the health and
safety of employees
at work, 1992

Mines, Works
and Minerals
Ordinances,
1968

Atomic Energy
and Radiation
Protection Act,
2005

Radiation
Protection and
Waste Disposal
Regulations,
2011



IMO Conventions through Ministry of Transport: Directorate of Maritime Affairs (DMA)

ISM CODE
(Internation
al Safety
Manageme
nt Code for
the safe
operations
of ships)

STCW (standard
of training of
seafarers and
certificcation of
watchkeepers)

ISPS CODE
(Internation
al ship and
port
security)

MARPOL
(Marine
Pollution
Prevention)

SOLAS
(Safety of
Lives at
Sea)

MARITIME
LABOUR
CONVENTIONS)



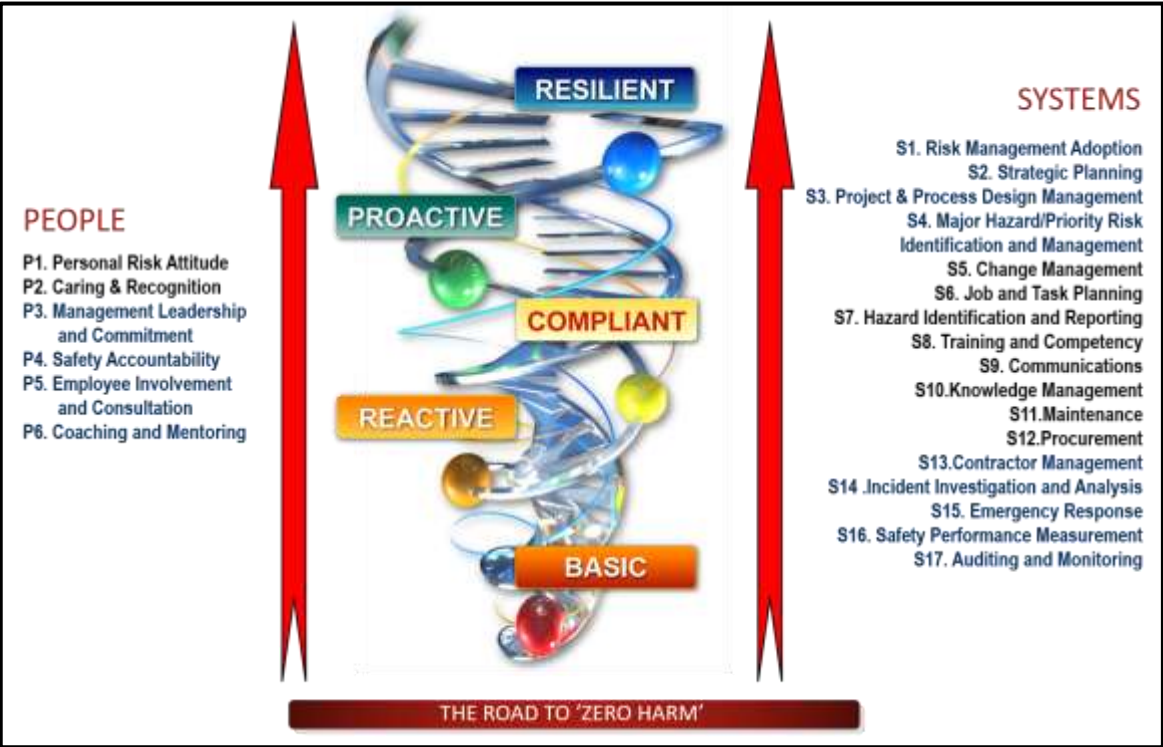


www.dedmaninamibia.com

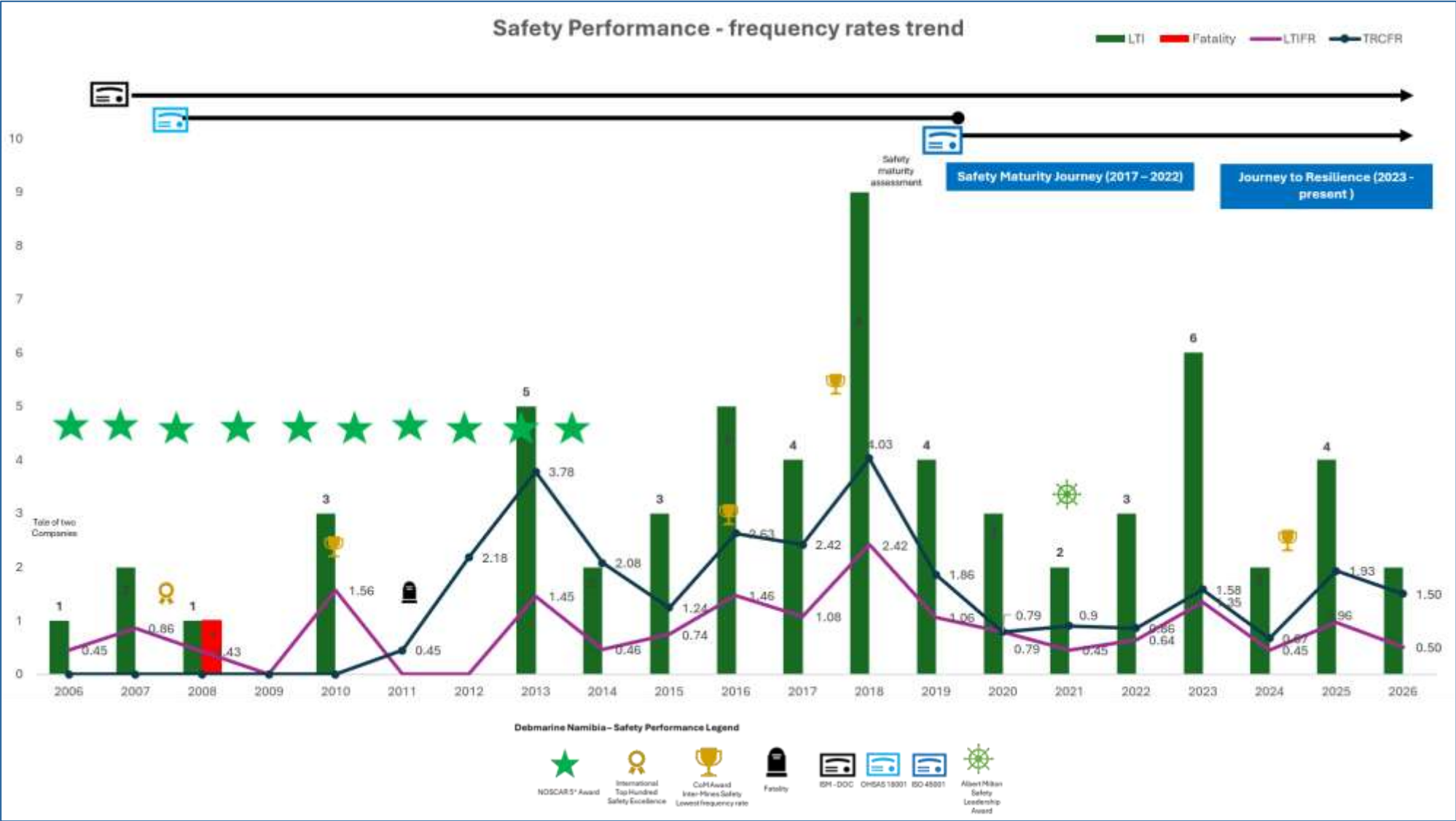


2010 THE INITIAL START...ZERO HARM..

- Fatal Risk Control Guidelines
- Focus on SHES Observations (Safety, Health, Environment and Security)
 - SHES focus on Low energy & Low impact risk and minor injuries
- Complaint for most part



PEOPLE CATEGORIESs	BASIC	REACTIVE	COMPLIANT	PROACTIVE	RESILIENT
P1. Personal Risk Attitude					
P2. Caring and Recognition					
P3. Management Leadership and Commitment					
P4. Safety Accountability					
P5. Employee Involvement and Consultation					
P6. Coaching and Mentoring					
SYSTEMS CATEGORIES					
S1. Risk Management Adoption					
S2. Strategic Planning					
S3. Project and Process Design Management					
S4. Major / Priority Risk Identification & Management					
S5. Change Management					
S6. Job and Task Planning					
S7. Hazard Identification and Reporting					
S8. Training and Competency					
S9. Communications					
S10. Knowledge Management					
S11. Maintenance					
S12. Procurement					
S13. Contractor Management					
S14. Incident Investigation and Analysis					
S15. Emergency Response					
S16. Performance Measurement					
S17. Monitoring and Auditing					





www.dedmaninamibia.com



2017 THE BEGINNING OF THE JOURNEY....

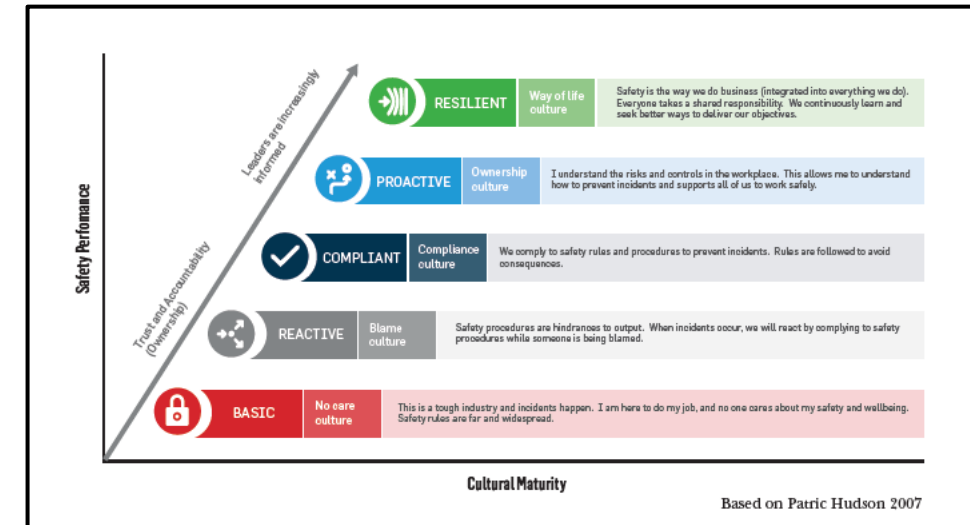
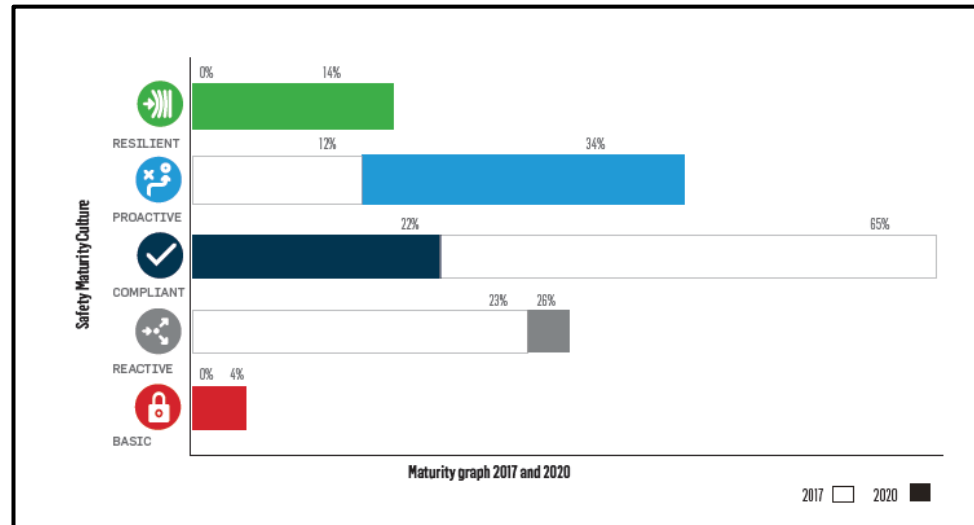
*In September 2017,
Debmarine Namibia
partnered with Hatch
and The Institute for
Telling Development
(ITD).*

ALL HANDS ON DECK!

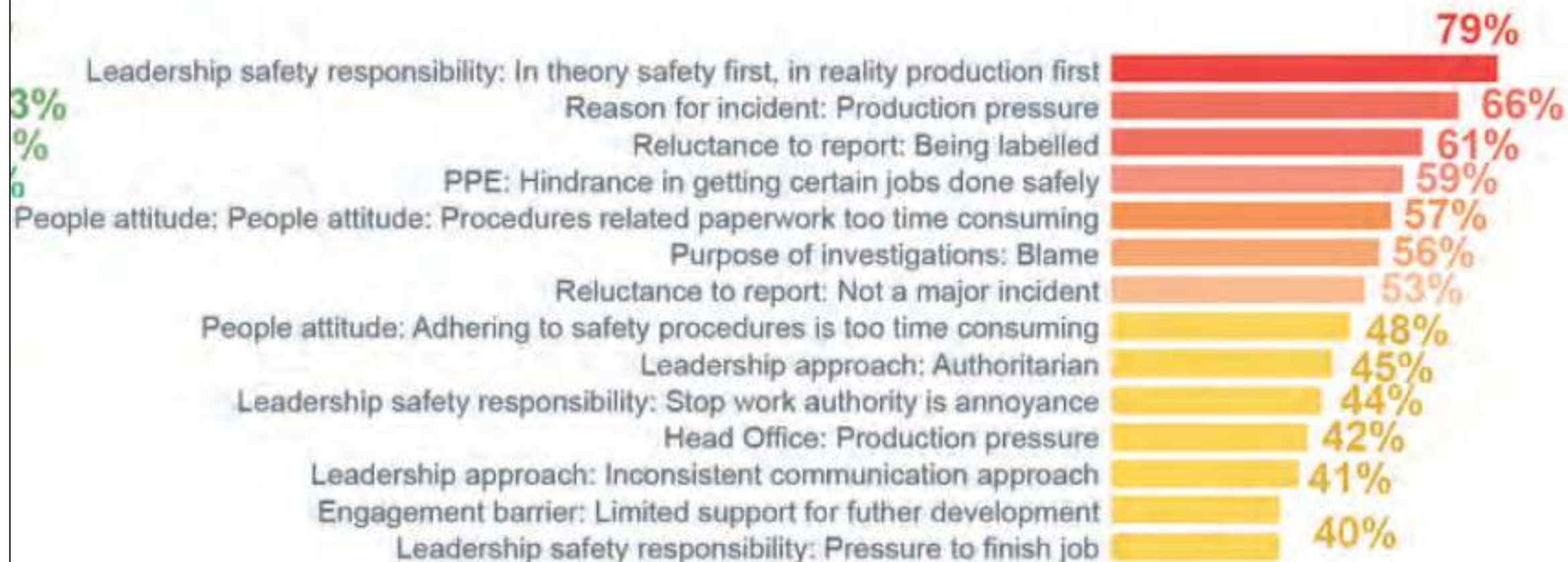


SAFETY MATURITY JOURNEY– INITIAL CULTURAL ASSESSMENT

- Production pressures
- Leaders not listening
- Ownership
- Accountability



DISABLERS





VALUES, MOTTOS AND THE LEADERSHIP LENSE

Creating a psychologically safe environment, no fear of speaking up, everyone is a safety leader

SAFETY MATURITY JOURNEY– INITIAL CULTURAL ASSESSMENT

DEBMARINE NAMIBIA VALUES

PUT SAFETY FIRST

TEAMWORK

INTEGRITY

EXCELLENCE

RESPECT

Values are a resource for proactive resilience to be guided towards doing what is right, unaffected by external pressure.

DEBMARINE NAMIBIA MOTTOS

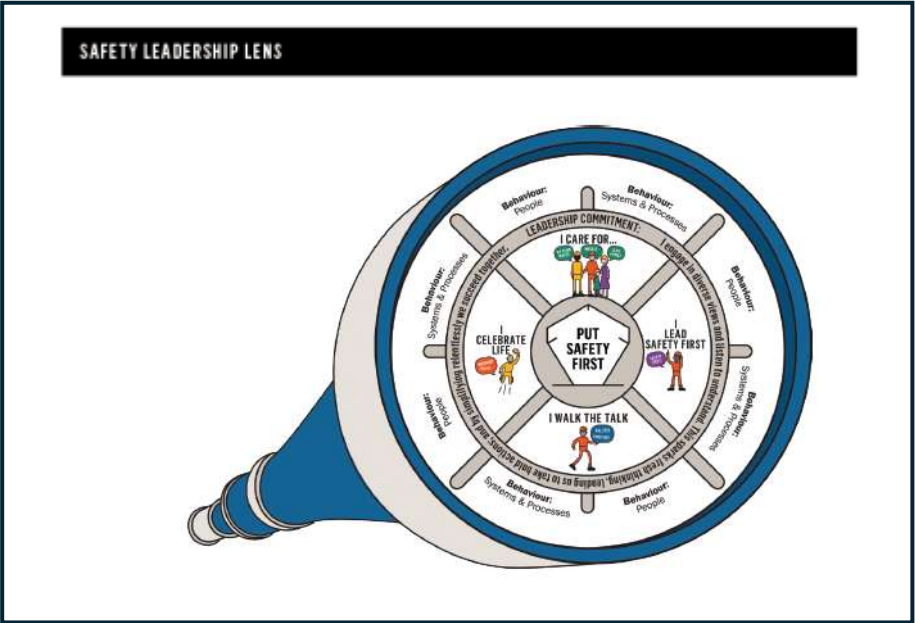
CARE FOR...

CELEBRATE LIFE

LEAD SAFETY FIRST

WALK THE TALK

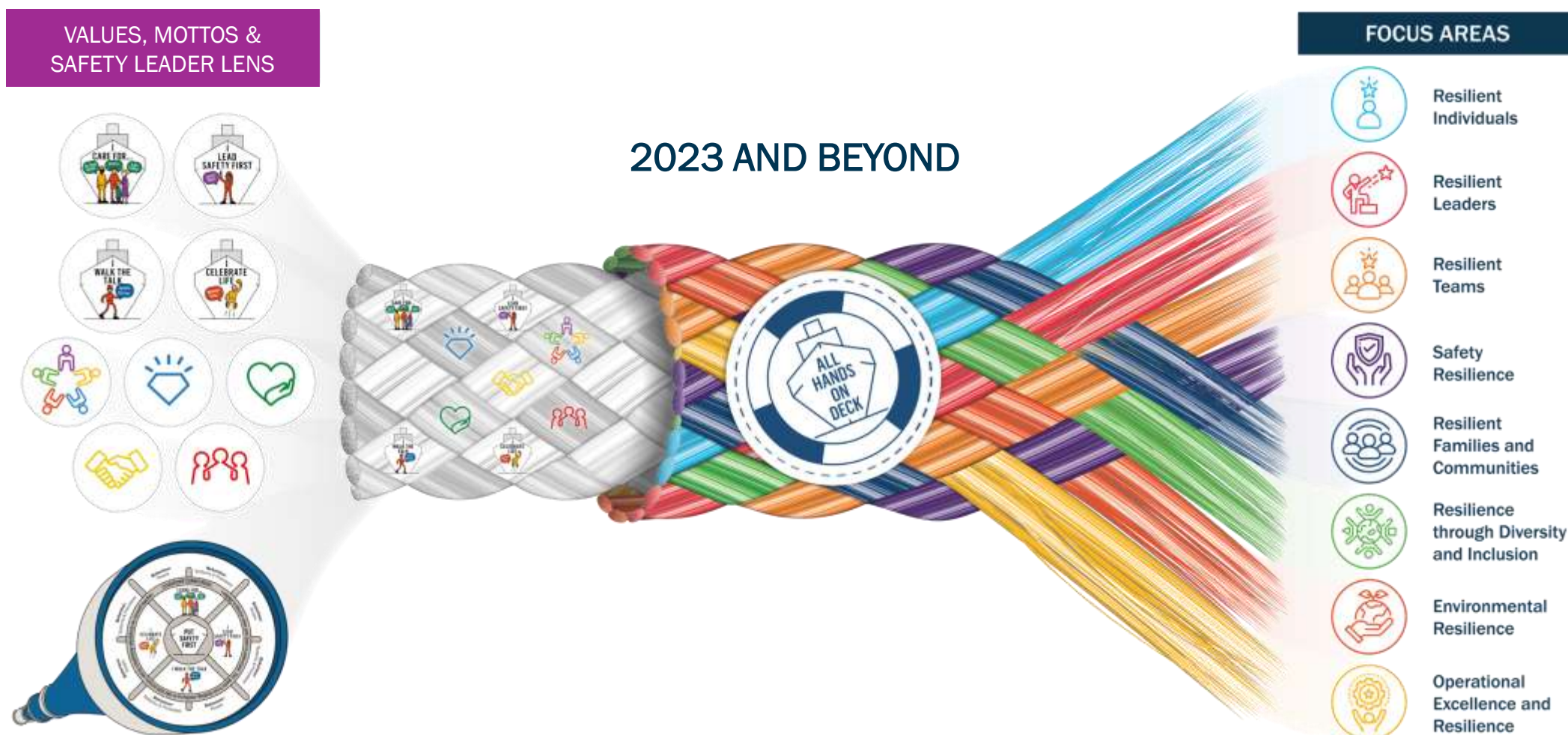
They pose the best probability for traction towards resilience by giving the Values ‘legs to walk on’ They energises!.



Safety leaders are those people who set a good example even when no one is watching and who influence and help others to do the right thing.

SAFETY MATURITY JOURNEY- INITIAL CULTURAL ASSESSMENT

integrated way towards organisational resilience



SAFETY MATURITY JOURNEY- SENSE MAKING THROUGH DIALOGUE



SAFETY MATURITY JOURNEY– INITIAL CULTURAL ASSESSMENT

- Psychological safe environment
- Change the language on production
- Everyone is a safety leader
- Around the table there are no titles
- Safety is an equalizer
- Each individual is at the center of the safety maturity journey





2018/2019 ESTABLISHING PRIORITY UNWANTED EVENTS

CRITICAL CONTROLS

Put the focus and energies on those high risk, high impact events that can result in the loss of life or life altering injuries

FLEET SAFETY– CRITICAL CONTROL MANAGEMENT

PUE No	CC 01	CC 02	CC 03	CC 04	CC 05	CC 06	CC 07	CC 08	CC 09	CC 10	CC 11	CC 12
PUE_07-Hazardous Materials	✓							1				
PUE_11-Confined Spaces	✓	✓	✓			✓		✓			✓	
PUE_30-Hotwork	✓	✓	✓	✓	✓	✓	✓	✓	✓			
PUE_37-Working @ Heights	✓	✓	✓	1		✓	✓	✓				
PUE_39-Lifting		✓			✓		✓	✓	✓	✓	✓	✓
PUE_40-Inadvertent contact with Live EI Energy	✓	1	1	✓	✓	2		2	1	✓		
PUE_82-Loss of Containment of Pressurised Pipes	✓	✓	✓	✓								
PUE_83-Loss of Containment from Pressure Vessel	✓	✓	✓	✓								
PUE_84-Loss of Sampling Capability	✓	✓	✓	✓	✓	✓	✓					
PUE_85-Fire in Namdeb Centre	✓	✓	✓	✓	✓	✓	✓	✓				
PUE_86-Launch & Recovery of FRC	✓	✓	✓	✓	✓		✓					
PUE_87-Structural Integrity	✓	✓	✓									
PUE_88-Exposure to Unguarded Moving Equipment	1	1	1	1	✓	✓						
PUE_89-Loss of Power	✓		1			1			1	✓		
PUE_90-Flooding of a Vessel at Sea	✓		✓	2			✓					
PUE_91-Loss of Stable Equilibrium	✓	✓	✓	✓	✓							
PUE_92-Refueling @ Sea		✓		✓	✓	✓						
PUE_93-Fixed Wing Aircraft	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
PUE_94-Helicopter Operations	✓	✓	✓		✓	✓						
PUE_95-Diving Operations	✓	✓				✓	✓	✓				
PUE_96-Loss of Control of Vessel Heat Sources												
PUE_97-Production Downtime Crawler Vessels	✓	2	✓		✓	1	✓					

Critical Control Codes
✓ = Active Critical Control for PUE
ARC = Action Required - Action Required
QMA = Ongoing Monitoring Action
QAF = Corrective Action In Progress
NOT = Control Noted

- Ownership is with the line management
- There are only 3 people in the safety department
- There are champions and owners for each PUE
- We celebrate the red
- We question too much green
- SAP systems provide feedback in terms of notifications
- There is a 24hr check on our Critical Controls
- Weekly dashboard reviews

Work Execution Document				
BASED ON		JRA-V402-ENG-00004-1.9-2025.11.28		
JOB CARD NUMBER		DATE	2025.11.28	
TASK NAME		DA Bin Liner Maintenance (ENG)		
Top Events: PUE's		PUE_30 Inadvertent Ignition of Combustible Material Due to Hot Work PUE_37 People, Tools, Parts or Equipment falling from Elevated Platforms PUE_39 Inadvertent Dropping / Swinging Loads onto PUE_40 Inadvertent contact with live electrical energy PUE_88 Exposure to Unguarded Moving Equipment		
Applicable procedure:		-		
Required Tools:		Battery Impact Guns, Cutting torches, Welding cables, ropes, crowbar,spikes, hand tools		
Impact on people/work:		No work on spout or movement of hoist possible		
TASK STEPS		NO-GO	COMPLIANT	NON COMPLIANT
1	Opening of hatches to launder and entering of launder/DA bin			
1. Have you inspected the Launder/DA bin for rocks?				
2. Have you locked out the mining air system valve?		x		
3. Have you isolated the main hoist and locked out the valve?		x		
4. Have you followed the lock out standard?		x		
5. Have you completed the lock out check list?		x		
2	Removal off old liners			
1. Have you completed a hotwork				
2. Is a steady stream of water in the vicinity during hotwork?				

- Empower teams to exercise stop work authority
- Establish Job Risk Assessment
- Identify the Critical Controls
- No-GO is the stop work authority
- Work Execution Documents when executing work
- The HOD is responsible for the WED's



UNACCEPTABLE BEHAVIOR

Taking ownership and holding people accountable

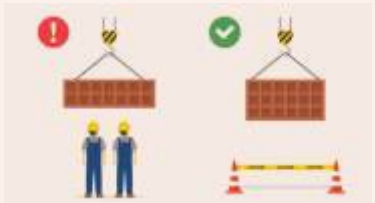
There are consequences to disregarding the safety cardinal rules

UNACCEPTABLE BEHAVIOUR

We don't gamble with each others' lives



Walking through **barriers** without permission.



Walking under **suspended loads**.



Overloading lifting equipment (cranes, chain blocks, slings).



Not **isolating** equipment/ energies including upstream & downstream.



Working on **live equipment** without risk assessment/ authorisation.



Not **hooking** on whilst W@H.



Doing hot work without a **Hot Work Permit** and a **Fire watch**.



Leaving **safety guards** off machinery that is not locked out.



Entering an **enclosed/ confined space** without a permit



Tampering with **LSA & FFE**.



Conducting critical work without a **JSA/SLAM/WRAC**.



Bypassing safety systems / interlocks.



Leaving the **galley unattended** whilst the stove and other cooking appliances are switched on.



www.dabmarinehamibia.com



SIMPLIFICATION OF SYSTEMS WITH HATCH AFRICA

Radical simplification: 513 documents became 44

513

SHE MS documents
before review (2018)



44

1 SHE manual · 29 standards
6 core procedures · 8 guidelines

How it was done (2018–2019)

- Co-designed with employees at every level, not imposed from a head-office template
- Every document categorised and reviewed against a simple framework-integration model
- Champions selected across the organisation to carry the standards reviews
- All users retrained on the consolidated system and simplified document control

“The fastest way to get killed on a manned space flight is to not follow standard operating procedure. The second quickest way is to always follow standard operating procedure.” — Astronaut Karol ‘Bo’ Bobko



2019 ELIMINATION OF FATALITIES WITH ANGLO AMERICAN

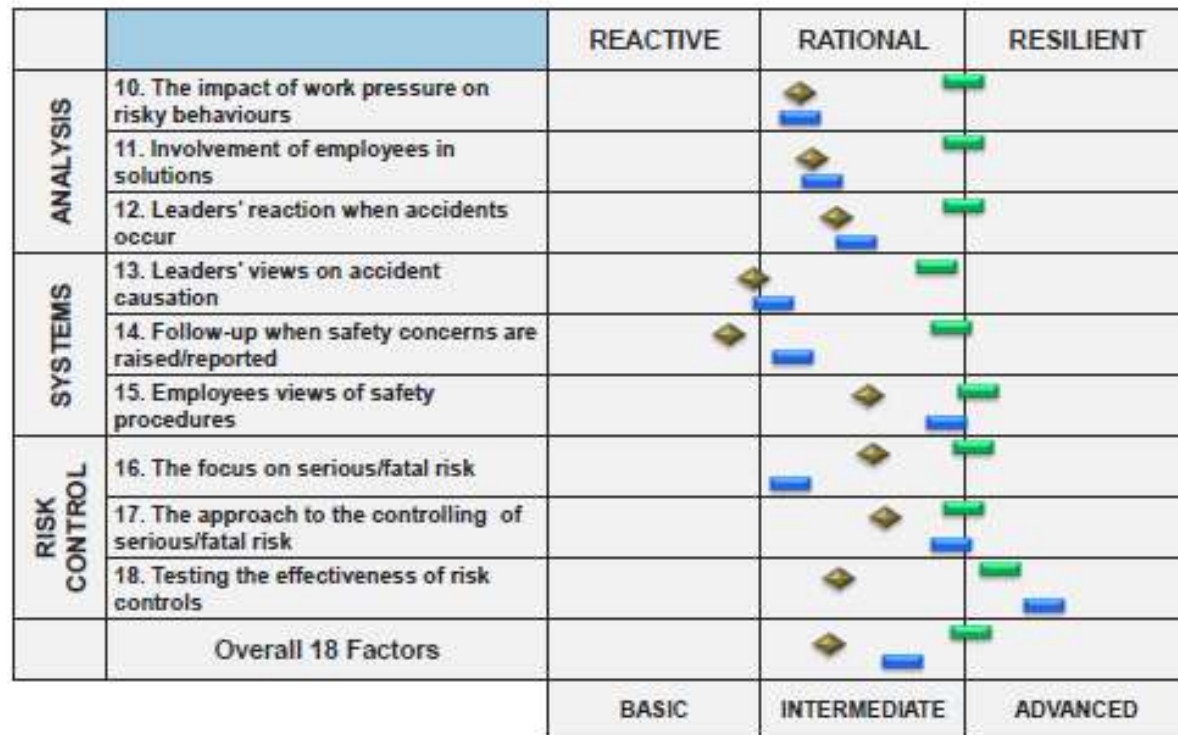
To fully understand the causation that leads to fatalities in Anglo American

INTERNATIONAL BENCHMARK		
28 'Top Companies' from 400+ companies		
Type of industry	Number	Location
Aviation/Defense	1	Global
Power utilities	8	USA
Oil and gas	2	Global
Project engineering	3	Global
Construction	3	Global
Mining	4	Australia/USA
Mineral Processing	3	Australia/Europe
Manufacturing	2	Global
Renewable energy	1	USA
Utility Services	2	USA

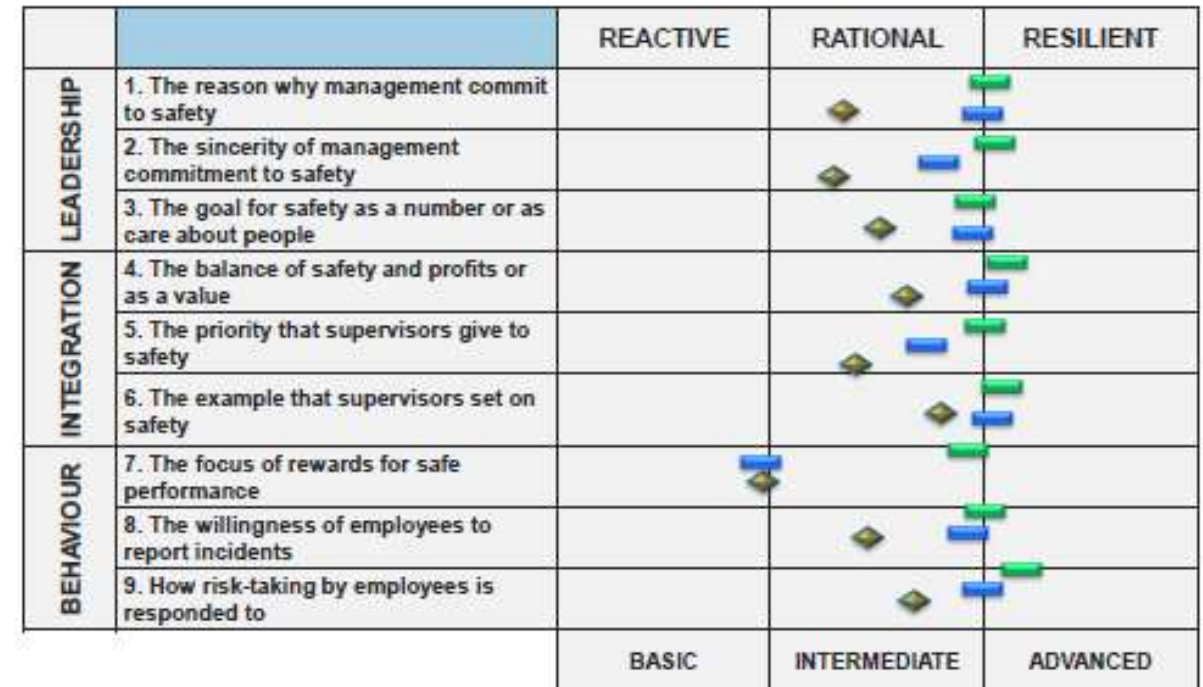
PLATINUM	1	Amandelbult Mine
	2	Waterval Smelter
	3	RBMR
	4	ACP
	5	PMR
	6	Amandelbult Plant
	7	Polokwane Smelter
	8	Mortimer
	9	Mogalakwena
COAL SA	10	Isibonelo
	11	Goedehoop
	12	Khwezela
	13	Greenside
	14	Zibulo
	15	Mafube
DEBEERS	16	Venetia
KUMBA	17	Sishen
	18	Kolomela
DEBSWANA	19	Jwaneng
	20	Orapa

ELIMINATION OF FATALITIES MARCH 2019

Culture Profile - Summary



Culture Profile - Summary





THE HUNTING FOR RISKS WITH THE SAFE SENTRIES: SAFEMAP

We adopted new practical safety risk identification methods.
We use SafeSENTRY as a rotational lookout role for colleagues to spot behaviours, conditions
and work that we might otherwise overlook while working.

safeSENTRY: born in the EOF, raised in Debmarine

A team-based practice built on a rotating ‘safe sentry’ — a quasi-formal, authorised leadership role. The sentry of the day walks the workplaces several times, focused exclusively on risk discovery: short observations, a fresh set of eyes on work in progress, direct dialogue with supervisors and managers.



Roving eyes for dynamic risk

Risks constantly shift in likelihood, exposure and impact. Multiple rotating observers see what those inside the work have stopped noticing.



Transparency defeats risk secrecy

Continuous, open feedback about risk on the team gradually builds willingness to report — reshaping culture towards resilience.

Embedded to last

- Demand outgrew supply: untrained crew asked when their turn for training would come
- Master safeSENTRY course created — 4–6 champions per vessel train and coach at the front line
- Now part of standard onboarding: every new employee is trained within their first year

“In Debmarine Namibia, everyone is a safety leader — the sentry role ensures everyone gets a turn to lead safe practice, in an authorised, rotation-based role.”



Crawler pilot Cedric Swarts performing a safesentry on the crawler deck after maintenance.



Maintenance assistant Sam Lukonga ensuring all critical controls during a hotwork task is active.



www.deltamarineamibia.com



CONTINUOUS EVOLVING WITH DELTA COMMANDERS

We adopted new practical safety risk identification methods. We use DELTA to identify deep latent triggers for accidents in Critical Controls.

IMPROVED PRACTICE · RISK DISCOVERY

Deep Elimination of Latent Triggers of Accidents. The premise: the trigger of the next fatality or catastrophe possibly already exists — in hazards, practices, processes, systems, and above all in culture. A system is only as good as how often it is effectively challenged.



“Strike” teams

A trained cross-section of employees and leaders, randomly assigned across the fleet



Assess in the field

Gather intelligence on the risk, go to the work, and test critical controls in real time as work happens

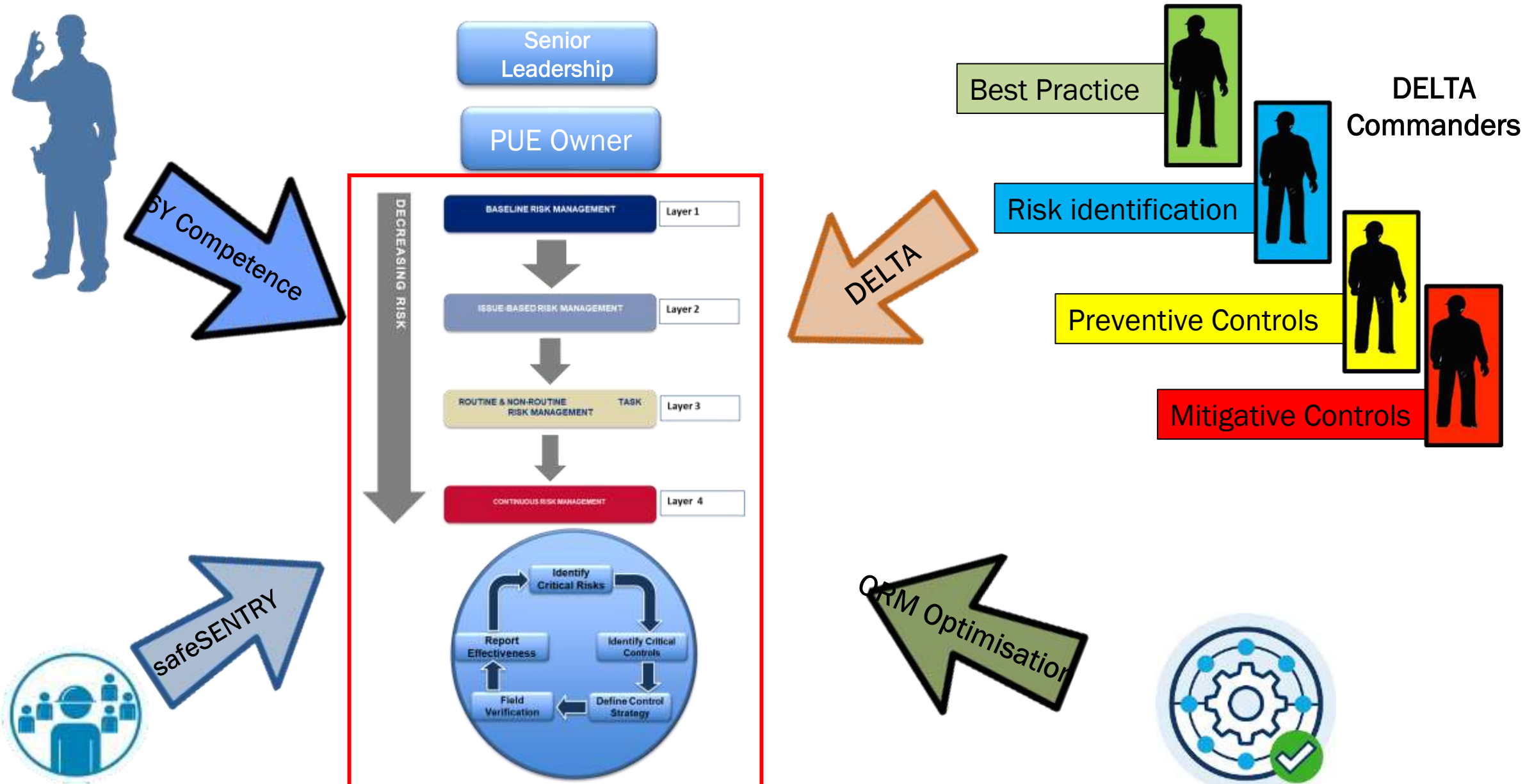


Review annually

Critical-risk results reviewed each year; controls updated and integrated back into the management system

ANNUAL FIELD ASSESSMENTS TARGET THE PRIORITY UNWANTED EVENTS (PUEs)

Hazardous materials · Confined spaces · Hot work · Working at heights · Lifting · Isolation · Helicopter operations · Flooding of a vessel at sea · Loss of stable equilibrium · Refuelling at sea





www.deltamarinenamibia.com



2021 BENGUELA GEM DESIGN FOR SAFETY

With more than 1300 systems and subsystems, the AMV3 has followed a safety centric approach to ensure maximum compliance to safety standards.

BENGUELA GEM DESIGN FOR SAFETY

How do we ensure the delivery of a safe asset into operations?

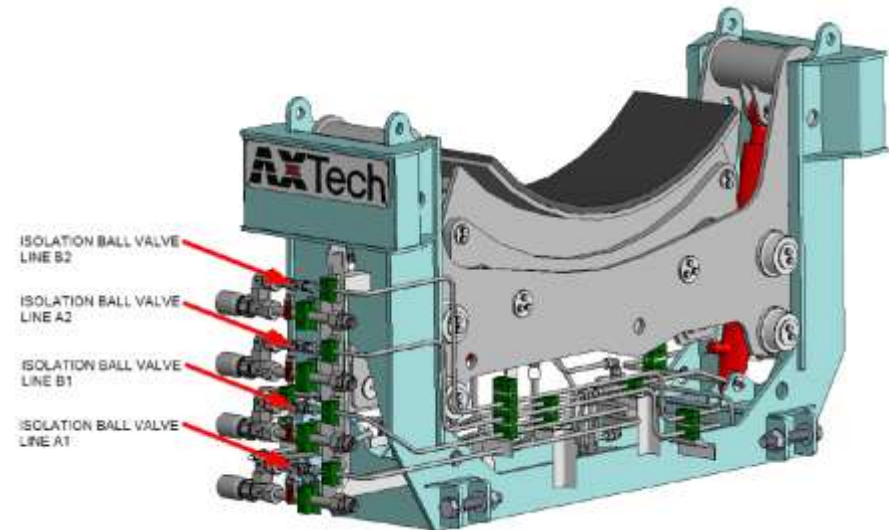
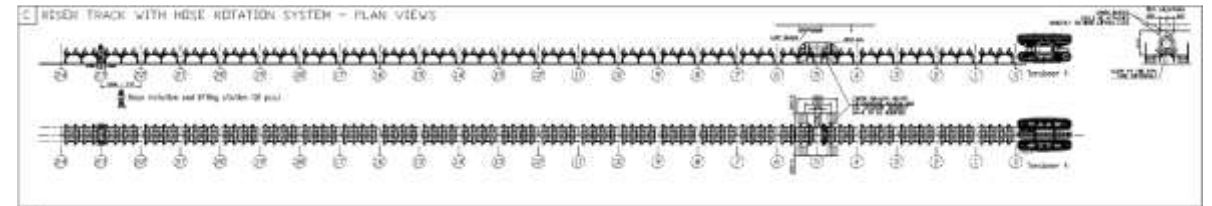
The objective of behind Design for Safety is to ensure that Safety Standards and Industry Best Practices are observed when systems and equipment are being designed, thus developing inherently safe equipment and systems.



Subsequently, equipment/systems that are designed whilst observing safety standards are generally safer to operate when compared with equipment/systems that are retrofitted to comply with standards and regulations.



Hands-free semi-automated Hose
Handling System



Hands-free semi-automated Hose
Rotation System



All the moving parts on the ball mill are guarded.
Ball mills are fitted with lockable access.



The gate-type guards fitted on the sinks scrubber have switches that disable the scrubber when the gate is open.



Extensive maintainability and operability reviews have led to the installation of several strategically poisoned crawl beams. This minimises the various risks associated with rigging and lifting operations



Wherever practicable, crawl beams have been extended to stair cases or have designated platforms allowing access to beam crawls.
This substantially reduces the risk of working at heights





www.debmarineamibia.com



DEBMARINE NAMIBIA SAFETY MATURITY JOURNEY

OUR JOURNEY to resilience



2017

- Okahandja Team Engagements
- Co-create Mottos
- 6 Activations

- Safety Maturity Assessment
- CEO pledge: Put Safety First
- Fatal risks and standards review

2018



- Leadership competency framework
- Leadership assessments
- 3 Activations
- Safe Minds & Delta Commander training



2019

- Safety Maturity Re-assessment
- 8 Activations
- Final Safety standards and SHE SMS manual

2020



2021

- Covid-19 support and mental health
- 7 Activations
- Master Safe Sentry training

2022



- Stolzenfeld Resilient Mindset team engagements

OUR JOURNEY to resilience



2023

- Interim assessment results
- 6 Activations

- Journey to Resilience (rope)
- Interim maturity assessment
- 2 Activations



2024



2025

- Maturity re-assessment
- Cost resilience focus
- Values refresh
- 5 Activations

- Organisational resilience focus
- Assessment feedback & way forward



2026

Everyone is a Safety Leader

SUSTAINING CULTURE GAINS TOWARDS RESILIENCE

Building a resilient safety culture through our values, mottos and Journey to Resilience.



LEADER ENABLEMENT

- Application of culture-specific leadership competency framework. (20 key competencies).
- Individual and Group Coaching.



TEAM ENGAGEMENT

Total Debmarmine Namibia team engagements:

- 2018 Values Sessions
- 2022 Resilient Mindset and onboarding new employees.



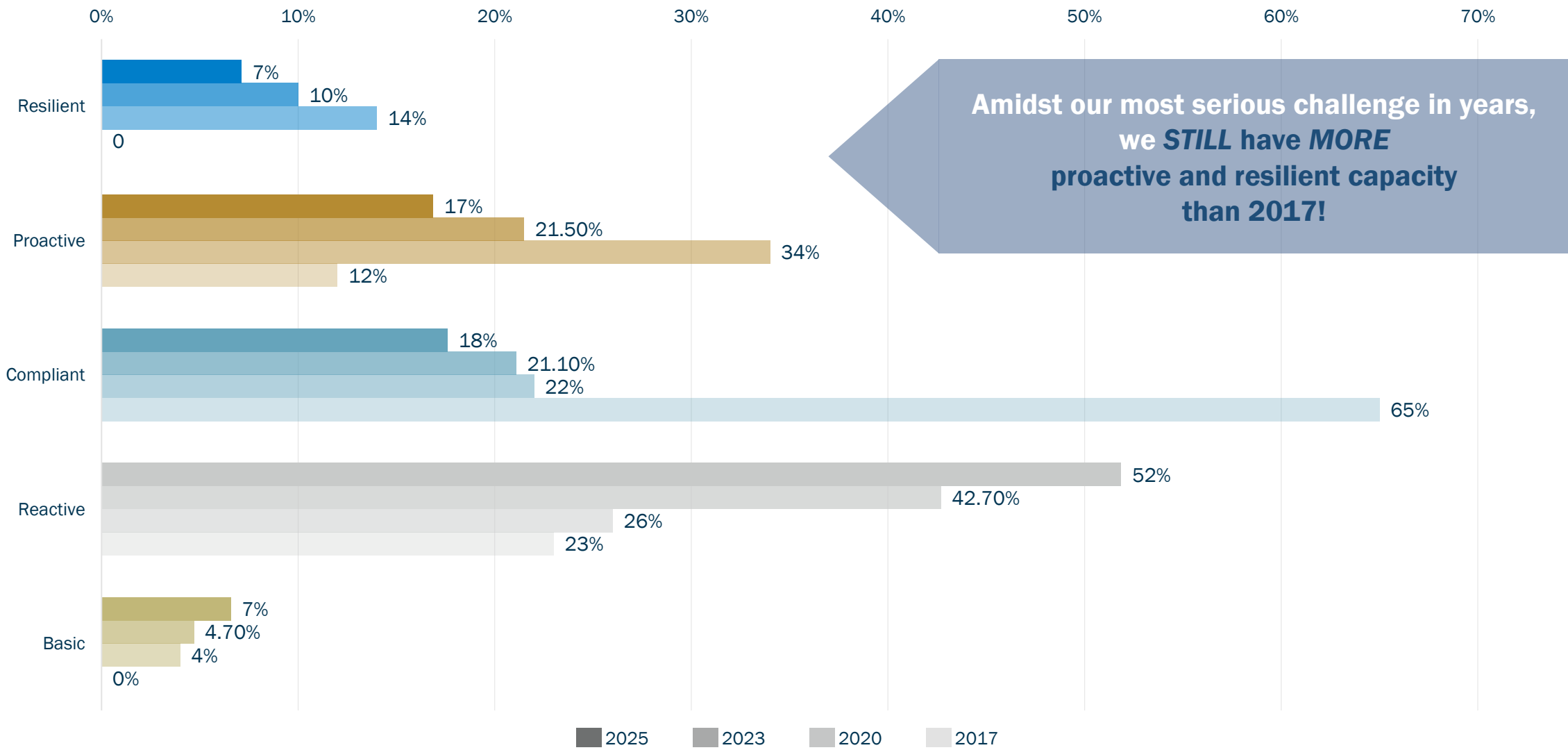
ACTIVATIONS

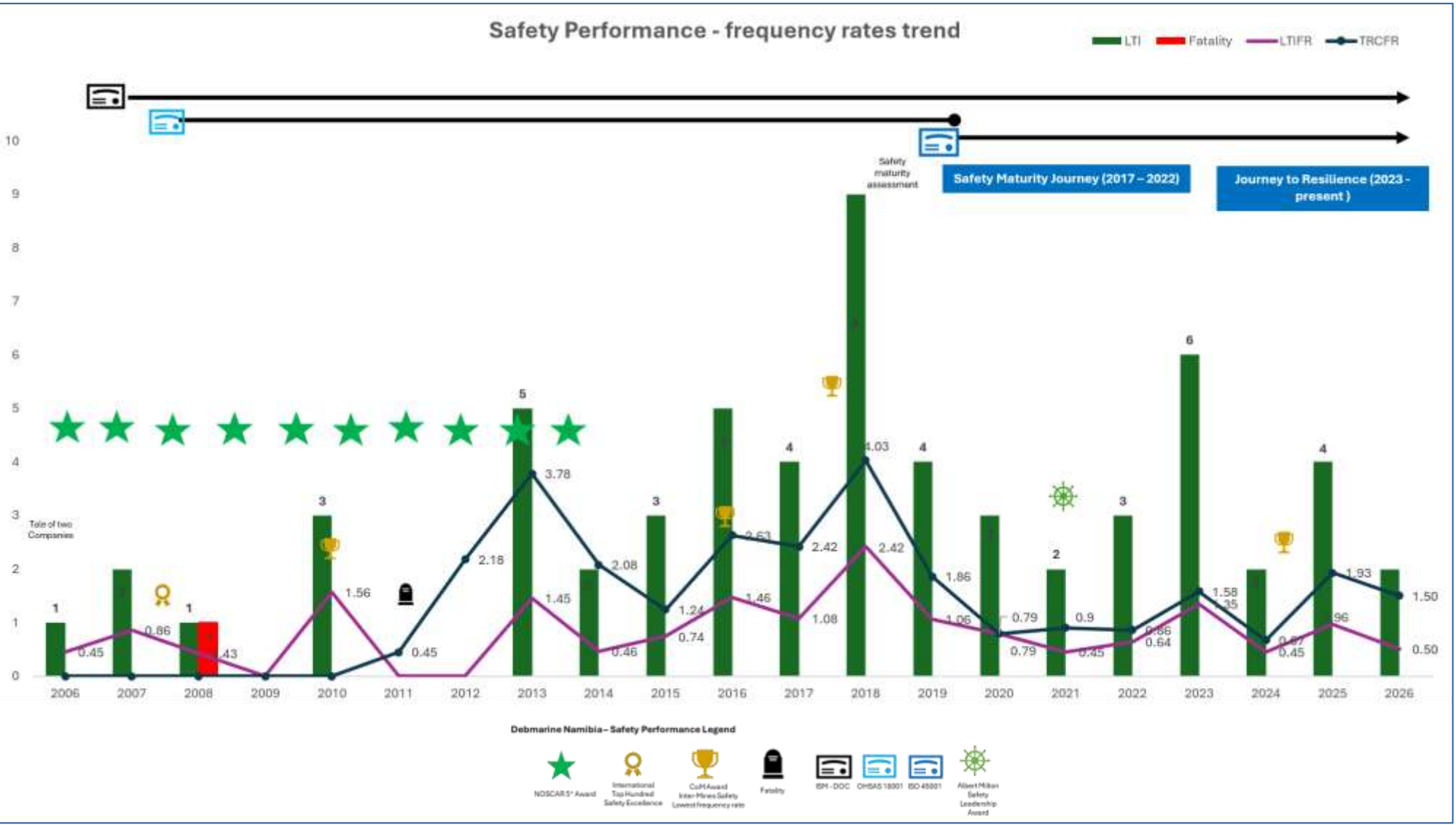
Continuous leader-led team conversation topics applying:

- Values
- Mottos



MATURITY OVER TIME | 2017 - 2025



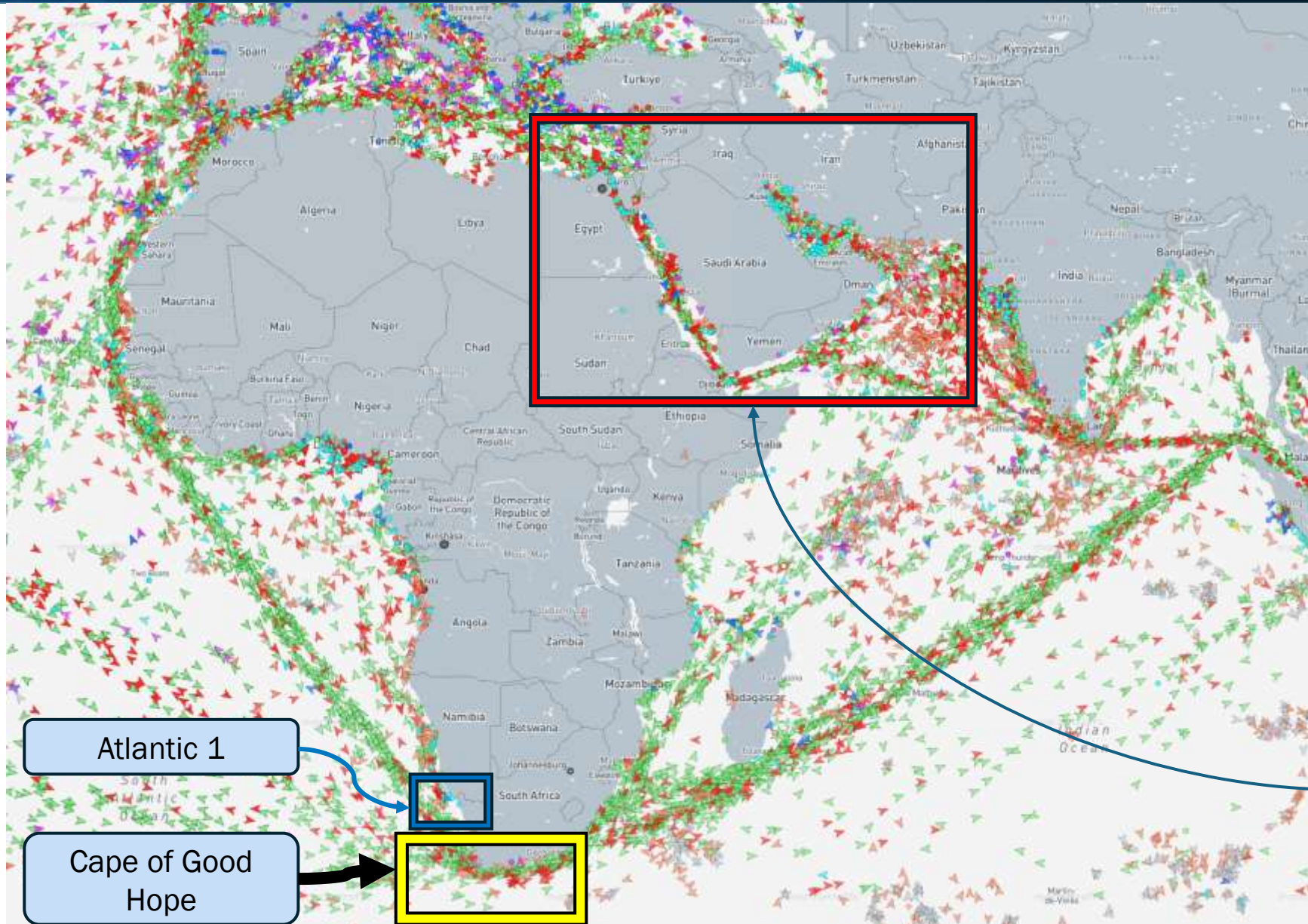




IMPACT OF US/IRAN WAR ON MARINE TRAFFIC

- The human is the strongest link
- Systems must be agile
- Readiness to respond is key
- Psychological safety is a requirement
- The journey does not have an end

RISK ARE DYNAMIC THE HUMAN IS THE STRONGEST LINK



- ✓ Vessels are choosing the Cape of Good Hope to avoid security risks, such as vessel attacks and hijackings, in the middle east.
- ✓ Rerouting via Cape of Good Hope is longer and add 3000 to 4000 NM to the voyage.
- ✓ The Cape of Good Hope route provides a safe passage for the crew and avoids high war-risk insurance premiums, and offers more predictable schedules than navigating volatile conflict zones

Shorter route through the Suez Canal is being avoided due to the conflict in the middle east.

INCREASE IN MARINE TRAFFIC – BG COLLISION COURSE

